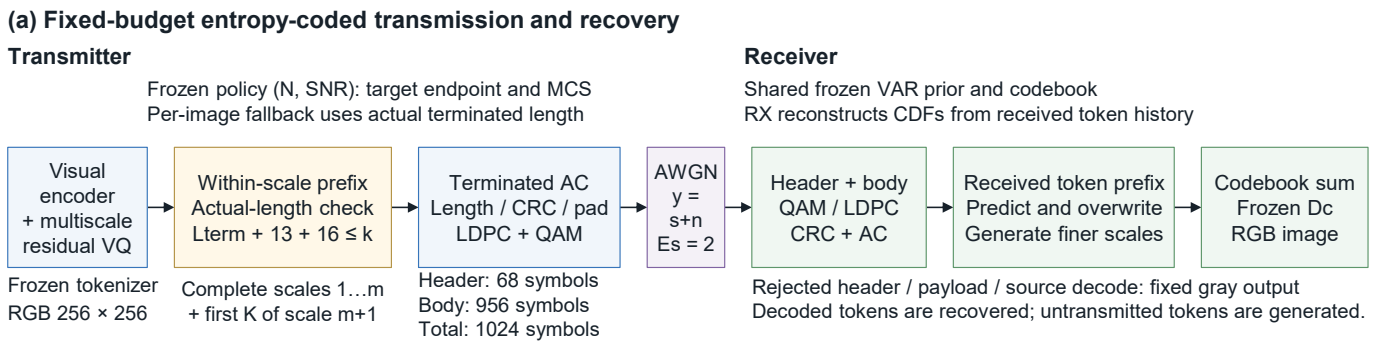

(a) Fixed-budget entropy-coded transmission and recovery
Transmitter
Frozen policy (N, SNR): target endpoint and MCS
Per-image fallback uses actual terminated length
Receiver
Shared frozen VAR prior and codebook
RX reconstructs CDFs from received token history
Visual encoder + multiscale residual VQ
Within-scale prefix Actual-length check Lterm + 13 + 16 ≤ k
Terminated AC Length / CRC / pad LDPC + QAM
AWGN y = s+n Es = 2
Header + body QAM / LDPC CRC + AC
Received token prefix Predict and overwrite Generate finer scales
Codebook sum Frozen Dc RGB image
Frozen tokenizer RGB 256 × 256
Complete scales 1…m + first K of scale m+1
Header: 68 symbols
Body: 956 symbols
Total: 1024 symbols
Rejected header / payload / source decode: fixed gray output
Decoded tokens are recovered; untransmitted tokens are generated.
Entropy-bitstream capacity utilization · 10 dB · common500
Terminated entropy bits / (k − 13 length bits − 16 CRC bits); usable capacity = 2520 bits
WHOLE
77.70%
PARTIAL
95.21%
(b) Prefix-constrained reconstruction · example m = 7, K = 75
Decoded scales 1–7 155 token indices
Predict all 100 tokens of boundary scale 8 from decoded context
Overwrite first 75 with received tokens Keep 25 predictions
Advance with the merged scale-8 state (75 received + 25 predicted)
Generate finer scales Scale 9: 169 tokens Scale 10: 256 tokens
Raster order fixes the received positions. Only received tokens overwrite predictions; source truth is unavailable at RX.